Ex.No: 11	Data Modelling and Analytics with Pivot Table in Power BI
Date:	

AIM:

To make a pivot table for data modelling and analytics using power bi.

PROCEDURE:

- **Key Performance Indicators (KPIs) - Card Visuals**

  - Total Revenue → SUM (Orders [Revenue])
  - Total Profit → [total_revenue] - [total_cost]
  - Profit Margin (%) → DIVIDE([total_profit], [total_revenue], 0)
  - Total Orders → DISTINCTCOUNT (Orders [Order ID])
  - Average Order Value (AOV) 🏷 → DIVIDE([total_revenue], [total_orders], 0)

Visualization: Use Card visuals for each KPI.

- **Sales Performance Over Time - Line Chart**

Measure:

  - Monthly Sales Trend → SUMMARIZE (Orders, FORMAT (Orders [Date], "YYYY-MM"), "Monthly Revenue", SUM (Orders [Revenue]))

Visualization: Use a Line Chart with:

  - X-axis → Month-Year (YYYY-MM)
  - Y-axis → Monthly Revenue

- **Top 5 Best-Selling Cookies - Bar Chart**

Measure:

  - Top-Selling Cookies → TOPN(5, SUMMARIZE(Orders, Orders[Product],"Total Sales", SUM(Orders[Units Sold])), [Total Sales], DESC)

Visualization: Use a Clustered Bar Chart with:

  - X-axis → Total Sales
  - Y-axis → Product Name (Descending Order)

- **Customer Insights - Table or Matrix**

Measures:

  - Customer-wise Revenue → CALCULATE([total_revenue], ALLEXCEPT (Orders, Orders [Customer ID]))
  - Total Orders per Customer → CALCULATE([total_orders], ALLEXCEPT (Orders, Orders [Customer ID]))
  - Top 5 Customers by Revenue → TOPN (5, SUMMARIZE (Orders, Orders [Customer ID], "Total Revenue", SUM (Orders [Revenue])),

Visualization: Use a Matrix or Table visual showing:

- Columns: Customer Name, Total Revenue, Total Orders

- **Profitability by Cookie Type - Pie Chart**

Measure:

- Profit per Cookie Type → SUMMARIZE(Orders, Orders[Product], "Total Profit", SUM(Orders[Revenue]) - SUM(Orders[Cost]))

Visualization: Use a Pie Chart with:

- Legend → Product Name
- Values → Total Profit

OUTPUT:

RESULT:

Thus pivot table for data modelling and analytics are implemented using power bi has been verified.